

Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC

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Abstract Leave-one-out cross-validation (LOO) and the widely applicable information criterion (WAIC) are methods for estimating pointwise out-of-sample prediction accuracy from a fitted Bayesian model using the log-likelihood evaluated at the posterior simulations of the parameter values. LOO and WAIC have various advantages over simpler estimates of predictive error such as AIC and DIC but are less used in practice because they involve additional computational steps. Here we lay out fast and stable computations for LOO and WAIC that can be performed using existing simulation draws. We introduce an efficient computation of LOO using Pareto-smoothed importance sampling (PSIS), a new procedure for regularizing importance weights. Although WAIC is asymptotically equal to LOO, we demonstrate that PSIS-LOO is more robust in the finite case with weak priors or influential observations. As a byproduct of our calculations, we also obtain approximate standard errors for estimated predictive errors and for comparison of predictive errors between two models. We implement the computations in an R package called `loo` and demonstrate using models fit with the Bayesian inference package Stan.

Keywords Bayesian computation · Leave-one-out cross-validation (LOO) · K -fold cross-validation · Widely applicable information criterion (WAIC) · Stan · Pareto smoothed importance sampling (PSIS)

1 Introduction

After fitting a Bayesian model we often want to measure its predictive accuracy, for its own sake or for purposes of model comparison, selection, or averaging (Geisser and Eddy 1979; Hoeting et al. 1999; Vehtari and Lampinen 2002; Ando and Tsay 2010; Vehtari and Ojanen 2012). Cross-validation and information criteria are two approaches to estimating out-of-sample predictive accuracy using within-sample fits (Akaike 1973; Stone 1977). In this article we consider computations using the log-likelihood evaluated at the usual posterior simulations of the parameters. Computation time for the predictive accuracy measures should be negligible compared to the cost of fitting the model and obtaining posterior draws in the first place.

Exact cross-validation requires re-fitting the model with different training sets. Approximate leave-one-out cross-validation (LOO) can be computed easily using importance sampling (IS; Gelfand et al. 1992; Gelfand 1996) but the resulting estimate is noisy, as the variance of the importance weights can be large or even infinite (Peruggia 1997; Epifani et al. 2008). Here we propose to use *Pareto smoothed importance sampling* (PSIS), a new approach that provides a more accurate and reliable estimate by fitting a Pareto distribution to the upper tail of the distribution of the importance weights. PSIS allows us to compute LOO using importance weights that would otherwise be unstable.

The widely applicable or Watanabe-Akaike information criterion (WAIC; Watanabe 2010) can be viewed as an

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improvement on the deviance information criterion (DIC) for Bayesian models. DIC has gained popularity in recent years, in part through its implementation in the graphical modeling package BUGS (Spiegelhalter et al. 2002; Spiegelhalter et al. 1994, 2003), but it is known to have some problems, which arise in part from not being fully Bayesian in that it is based on a point estimate (van der Linde 2005; Plummer 2008). For example, DIC can produce negative estimates of the effective number of parameters in a model and it is not defined for singular models. WAIC is fully Bayesian in that it uses the entire posterior distribution, and it is asymptotically equal to Bayesian cross-validation. Unlike DIC, WAIC is invariant to parametrization and also works for singular models.

Although WAIC is asymptotically equal to LOO, we demonstrate that PSIS-LOO is more robust in the finite case with weak priors or influential observations. We provide diagnostics for both PSIS-LOO and WAIC which tell when these approximations are likely to have large errors and computationally more intensive methods such as K -fold cross-validation should be used. Fast and stable computation and diagnostics for PSIS-LOO allows safe use of this new method in routine statistical practice. As a byproduct of our calculations, we also obtain approximate standard errors for estimated predictive errors and for the comparison of predictive errors between two models.

We implement the computations in a package for R (R Core Team 2016) called `loo` (Vehtari et al. 2016a,b) and demonstrate using models fit with the Bayesian inference package Stan (Stan Development Team 2016a,b).¹ All the computations are fast compared to the typical time required to fit the model in the first place. Although the examples provided in this paper all use Stan, the `loo` package is independent of Stan and can be used with models estimated by other software packages or custom user-written algorithms.

2 Estimating out-of-sample pointwise predictive accuracy using posterior simulations

Consider data $y_1, \dots, y_n$, modeled as independent given parameters θ ; thus $p(y|\theta) = \prod_{i=1}^n p(y_i|\theta)$. This formulation also encompasses latent variable models with $p(y_i|f_i, \theta)$, where f_i are latent variables. Also suppose we have a prior distribution $p(\theta)$, thus yielding a posterior distribution $p(\theta|y)$ and a posterior predictive distribution $p(\tilde{y}|y) = \int p(\tilde{y}_i|\theta)p(\theta|y)d\theta$. To maintain comparability with the given dataset and to get easier interpretation of the differences in scale of effective number of parameters, we define

a measure of predictive accuracy for the n data points taken one at a time:

elpd = expected log pointwise predictive density

for a new dataset

$$= \sum_{i=1}^n \int p_t(\tilde{y}_i) \log p(\tilde{y}_i|y) d\tilde{y}_i, \quad (1)$$

where $p_t(\tilde{y}_i)$ is the distribution representing the true data-generating process for $\tilde{y}_i$. The $p_t(\tilde{y}_i)$'s are unknown, and we will use cross-validation or WAIC to approximate (1). In a regression, these distributions are also implicitly conditioned on any predictors in the model. See Vehtari and Ojanen (2012) for other approaches to approximating $p_t(\tilde{y}_i)$ and discussion of alternative prediction tasks.

Instead of the log predictive density $\log p(\tilde{y}_i|y)$, other utility (or cost) functions $u(p(\tilde{y}|y), \tilde{y})$ could be used, such as classification error. Here we take the log score as the default for evaluating the predictive density (Geisser and Eddy 1979; Bernardo and Smith 1994; Gneiting and Raftery 2007).

A helpful quantity in the analysis is

lpd = log pointwise predictive density

$$= \sum_{i=1}^n \log p(y_i|y) = \sum_{i=1}^n \log \int p(y_i|\theta) p(\theta|y) d\theta. \quad (2)$$

The lpd of observed data y is an overestimate of the elpd for future data (1). To compute the lpd in practice, we can evaluate the expectation using draws from $p_{\text{post}}(\theta)$, the usual posterior simulations, which we label θ^s , $s = 1, \dots, S$:

$\widehat{\text{lpd}}$ = computed log pointwise predictive density

$$= \sum_{i=1}^n \log \left(\frac{1}{S} \sum_{s=1}^S p(y_i|\theta^s) \right). \quad (3)$$

2.1 Leave-one-out cross-validation

The Bayesian LOO estimate of out-of-sample predictive fit is

$$\text{elpd}_{\text{loo}} = \sum_{i=1}^n \log p(y_i|y_{-i}), \quad (4)$$

where

$$p(y_i|y_{-i}) = \int p(y_i|\theta) p(\theta|y_{-i}) d\theta \quad (5)$$

is the leave-one-out predictive density given the data without the i th data point.

¹ The `loo` R package is available from CRAN and <https://github.com/stan-dev/loo>. The corresponding code for Matlab, Octave, and Python is available at <https://github.com/avehtari/PSIS>.

2.1.1 Raw importance sampling

As noted by [Gelfand et al. \(1992\)](#), if the n points are conditionally independent in the data model we can then evaluate (5) with draws θ^s from the full posterior $p(\theta|y)$ using importance ratios

$$r_i^s = \frac{1}{p(y_i|\theta^s)} \propto \frac{p(\theta^s|y_{-i})}{p(\theta^s|y)} \quad (6)$$

to get the importance sampling leave-one-out (IS-LOO) predictive distribution,

$$p(\tilde{y}_i|y_{-i}) \approx \frac{\sum_{s=1}^S r_i^s p(\tilde{y}_i|\theta^s)}{\sum_{s=1}^S r_i^s}. \quad (7)$$

Evaluating this LOO log predictive density at the held-out data point y_i , we get

$$p(y_i|y_{-i}) \approx \frac{1}{\frac{1}{S} \sum_{s=1}^S \frac{1}{p(y_i|\theta^s)}}. \quad (8)$$

However, the posterior $p(\theta|y)$ is likely to have a smaller variance and thinner tails than the leave-one-out distributions $p(\theta|y_{-i})$, and thus a direct use of (8) induces instability because the importance ratios can have high or infinite variance.

For simple models the variance of the importance weights may be computed analytically. The necessary and sufficient conditions for the variance of the case-deletion importance sampling weights to be finite for a Bayesian linear model are given by [Peruggia \(1997\)](#). [Epifani et al. \(2008\)](#) extend the analytical results to generalized linear models and non-linear Michaelis-Menten models. However, these conditions can not be computed analytically in general.

[Koopman et al. \(2009\)](#) propose to use the maximum likelihood fit of the generalized Pareto distribution to the upper tail of the distribution of the importance ratios and use the fitted parameters to form a test for whether the variance of the importance ratios is finite. If the hypothesis test suggests the variance is infinite then they abandon importance sampling.

2.1.2 Truncated importance sampling

[Ionides \(2008\)](#) proposes a modification of importance sampling where the raw importance ratios r^s are replaced by truncated weights

$$w^s = \min(r^s, \sqrt{S\bar{r}}), \quad (9)$$

where $\bar{r} = \frac{1}{S} \sum_{s=1}^S r^s$. [Ionides \(2008\)](#) proves that the variance of the truncated importance sampling weights is guaranteed to be finite, and provides theoretical and experimental results showing that truncation using the threshold

$\sqrt{S\bar{r}}$ gives an importance sampling estimate with a mean square error close to an estimate with a case specific optimal truncation level. The downside of the truncation is that it introduces a bias, which can be large as we demonstrate in our experiments.

2.1.3 Pareto smoothed importance sampling

We can improve the LOO estimate using Pareto smoothed importance sampling (PSIS; [Vehtari and Gelman 2015](#)), which applies a smoothing procedure to the importance weights. We briefly review the motivation and steps of PSIS here, before moving on to focus on the goals of using and evaluating predictive information criteria.

As noted above, the distribution of the importance weights used in LOO may have a long right tail. We use the empirical Bayes estimate of [Zhang and Stephens \(2009\)](#) to fit a generalized Pareto distribution to the tail (20 % largest importance ratios). By examining the shape parameter k of the fitted Pareto distribution, we are able to obtain sample based estimates of the existence of the moments ([Koopman et al. 2009](#)). This extends the diagnostic approach of [Peruggia \(1997\)](#) and [Epifani et al. \(2008\)](#) to be used routinely with IS-LOO for any model with a factorizing likelihood.

[Epifani et al. \(2008\)](#) show that when estimating the leave-one-out predictive density, the central limit theorem holds if the distribution of the weights has finite variance. These results can be extended via the generalized central limit theorem for stable distributions. Thus, even if the variance of the importance weight distribution is infinite, if the mean exists then the accuracy of the estimate improves as additional posterior draws are obtained.

When the tail of the weight distribution is long, a direct use of importance sampling is sensitive to one or few largest values. By fitting a generalized Pareto distribution to the upper tail of the importance weights, we smooth these values. The procedure goes as follows:

1. Fit the generalized Pareto distribution to the 20% largest importance ratios r_s as computed in (6). The computation is done separately for each held-out data point i . In simulation experiments with thousands and tens of thousands of draws, we have found that the fit is not sensitive to the specific cutoff value (for a consistent estimation, the proportion of the samples above the cutoff should get smaller when the number of draws increases).
2. Stabilize the importance ratios by replacing the M largest ratios by the expected values of the order statistics of the fitted generalized Pareto distribution

$$F^{-1}\left(\frac{z-1/2}{M}\right), \quad z = 1, \dots, M,$$

where M is the number of simulation draws used to fit the Pareto (in this case, $M = 0.2 S$) and F^{-1} is the inverse-CDF of the generalized Pareto distribution. Label these new weights as $\tilde{w}_i^s$ where, again, s indexes the simulation draws and i indexes the data points; thus, for each i there is a distinct vector of S weights.

3. To guarantee finite variance of the estimate, truncate each vector of weights at $S^{3/4} \bar{w}_i$, where $\bar{w}_i$ is the average of the S smoothed weights corresponding to the distribution holding out data point i . Finally, label these truncated weights as w_i^s .

The above steps must be performed for each data point i . The result is a vector of weights w_i^s , $s = 1, \dots, S$, for each i , which in general should be better behaved than the raw importance ratios r_i^s from which they are constructed.

The results can then be combined to compute the desired LOO estimates. The PSIS estimate of the LOO expected log pointwise predictive density is

$$\widehat{\text{elpd}}_{\text{psis-loo}} = \sum_{i=1}^n \log \left(\frac{\sum_{s=1}^S w_i^s p(y_i | \theta^s)}{\sum_{s=1}^S w_i^s} \right). \quad (10)$$

The estimated shape parameter $\hat{k}$ of the generalized Pareto distribution can be used to assess the reliability of the estimate:

- If $k < \frac{1}{2}$, the variance of the raw importance ratios is finite, the central limit theorem holds, and the estimate converges quickly.
- If k is between $\frac{1}{2}$ and 1, the variance of the raw importance ratios is infinite but the mean exists, the generalized central limit theorem for stable distributions holds, and the convergence of the estimate is slower. The variance of the PSIS estimate is finite but may be large.
- If $k > 1$, the variance and the mean of the raw ratios distribution do not exist. The variance of the PSIS estimate is finite but may be large.

If the estimated tail shape parameter $\hat{k}$ exceeds 0.5, the user should be warned, although in practice we have observed good performance for values of $\hat{k}$ up to 0.7. Even if the PSIS estimate has a finite variance, when $\hat{k}$ exceeds 0.7 the user should consider sampling directly from $p(\theta^s | y_{-i})$ for the problematic i , use K -fold cross-validation (see Sect. 2.3), or use a more robust model.

The additional computational cost of sampling directly from each $p(\theta^s | y_{-i})$ is approximately the same as sampling from the full posterior, but it is recommended if the number of problematic data points is not too high.

A more robust model may also help because importance sampling is less likely to work well if the marginal posterior

$p(\theta^s | y)$ and LOO posterior $p(\theta^s | y_{-i})$ are very different. This is more likely to happen with a non-robust model and highly influential observations. A robust model may reduce the sensitivity to one or several highly influential observations, as we show in the examples in Sect. 4.

2.2 WAIC

WAIC (Watanabe 2010) is an alternative approach to estimating the expected log pointwise predictive density and is defined as

$$\widehat{\text{elpd}}_{\text{waic}} = \widehat{\text{lpd}} - \widehat{p}_{\text{waic}}, \quad (11)$$

where $\widehat{p}_{\text{waic}}$ is the estimated effective number of parameters and is computed based on the definition²

$$p_{\text{waic}} = \sum_{i=1}^n \text{var}_{\text{post}}(\log p(y_i | \theta)), \quad (12)$$

which we can calculate using the posterior variance of the log predictive density for each data point y_i , that is, $V_{s=1}^S \log p(y_i | \theta^s)$, where $V_{s=1}^S$ represents the sample variance, $V_{s=1}^S a_s = \frac{1}{S-1} \sum_{s=1}^S (a_s - \bar{a})^2$. Summing over all the data points y_i gives a simulation-estimated effective number of parameters,

$$\widehat{p}_{\text{waic}} = \sum_{i=1}^n V_{s=1}^S (\log p(y_i | \theta^s)). \quad (13)$$

For DIC, there is a similar variance-based computation of the number of parameters that is notoriously unreliable, but the WAIC version is more stable because it computes the variance separately for each data point and then takes the sum; the summing yields stability.

The *effective number of parameters* $\widehat{p}_{\text{waic}}$ can be used as measure of complexity of the model, but it should not be over-interpreted, as the original goal is to estimate the difference between $\widehat{\text{lpd}}$ and $\widehat{\text{elpd}}$. As shown by Gelman et al. (2014) and demonstrated also in Sect. 4, in the case of a weak prior, $\widehat{p}_{\text{waic}}$ can severely underestimate the difference between $\widehat{\text{lpd}}$ and $\widehat{\text{elpd}}$. For $\widehat{p}_{\text{waic}}$ there is no similar theory as for the moments of the importance sampling weight distribution, but based on our simulation experiments it seems that $\widehat{p}_{\text{waic}}$ is unreliable if any of the terms $V_{s=1}^S \log p(y_i | \theta^s)$ exceeds 0.4.

The different behavior of LOO and WAIC seen in the experiments can be understood by comparing Taylor series approximations. By defining a generating function of functional cumulants,

² In Gelman et al. (2013), the variance-based p_{waic} defined here is called $p_{\text{waic}2}$. There is also a mean-based formula, $p_{\text{waic}1}$, which we do not use here.

$$F(\alpha) = \sum_{i=1}^n \log E_{\text{post}}(p(y_i|\theta)^\alpha), \quad (14)$$

and applying a Taylor expansion of $F(\alpha)$ around 0 with $\alpha = -1$ we obtain an expansion of lpd_{loo}

$$\begin{aligned} \text{elpd}_{\text{loo}} &= F'(0) - \frac{1}{2}F''(0) + \frac{1}{6}F^{(3)}(0) \\ &\quad - \sum_{i=4}^{\infty} \frac{(-1)^i F^{(i)}(0)}{i!}. \end{aligned} \quad (15)$$

From the definition of $F(\alpha)$ we get

$$\begin{aligned} F(0) &= 0 \\ F(1) &= \sum_{i=1}^n \log E_{\text{post}}(p(y_i|\theta)) \\ F'(0) &= \sum_{i=1}^n E_{\text{post}}(\log p(y_i|\theta)) \\ F''(0) &= \sum_{i=1}^n \text{var}_{\text{post}}(\log p(y_i|\theta)). \end{aligned} \quad (16)$$

Furthermore

$$\text{lpd} = F(1) = F'(0) + \frac{1}{2}F''(0) + \frac{1}{6}F^{(3)}(0) + \sum_{i=4}^{\infty} \frac{F^{(i)}(0)}{i!}, \quad (17)$$

and the expansion for WAIC is then

$$\begin{aligned} \text{WAIC} &= F(1) - F''(0) \\ &= F'(0) - \frac{1}{2}F''(0) + \frac{1}{6}F^{(3)}(0) + \sum_{i=4}^{\infty} \frac{F^{(i)}(0)}{i!}. \end{aligned} \quad (18)$$

The first three terms of the expansion of WAIC match the expansion of LOO, and the rest of the terms match the expansion of lpd . Watanabe (2010) argues that, asymptotically, the latter terms have negligible contribution and thus asymptotic equivalence with LOO is obtained. However, the error can be significant in the case of finite n and weak prior information as shown by Gelman et al. (2014), and demonstrated also in Sect. 4. If the higher order terms are not negligible, then WAIC is biased towards lpd . To reduce this bias it is possible to compute additional series terms, but computing higher moments using a finite posterior sample increases the variance of the estimate and, based on our experiments, it is more difficult to control the bias-variance tradeoff than in PSIS-LOO. WAIC's larger bias compared to LOO is also demonstrated by Vehtari et al. (2016a,b) in the case of Gaussian processes with distributional posterior approximations. In the experiments we also demonstrate that we can

use truncated IS-LOO with heavy truncation to obtain similar bias towards lpd and similar estimate variance as in WAIC.

2.3 K-fold cross-validation

In this paper we focus on leave-one-out cross-validation and WAIC, but, for statistical and computational reasons, it can make sense to cross-validate using $K \ll n$ hold-out sets. In some ways, K -fold cross-validation is simpler than leave-one-out cross-validation but in other ways it is not. K -fold cross-validation requires refitting the model K times which can be computationally expensive whereas approximative LOO methods, such as PSIS-LOO, require only one evaluation of the model.

If in PSIS-LOO $\hat{k} > 0.7$ for a few i we recommend sampling directly from each corresponding $p(\theta^s|y_{-i})$, but if there are more than K problematic i , then we recommend checking the results using K -fold cross-validation. Vehtari and Lampinen (2002) demonstrate cases where IS-LOO fails (according to effective sample size estimates instead of the $\hat{k}$ diagnostic proposed here) for a large number of i and K -fold-CV produces more reliable results.

In Bayesian K -fold cross-validation, the data are partitioned into K subsets y_k , for $k = 1, \dots, K$, and then the model is fit separately to each training set $y_{(-k)}$, thus yielding a posterior distribution $p_{\text{post}(-k)}(\theta) = p(\theta|y_{(-k)})$. If the number of partitions is small (a typical value in the literature is $K = 10$), it is not so costly to simply re-fit the model separately to each training set. To maintain consistency with LOO and WAIC, we define predictive accuracy for each data point, so that the log predictive density for y_i , if it is in subset k , is

$$\log p(y_i|y_{(-k)}) = \log \int p(y_i|\theta)p(\theta|y_{(-k)})d\theta, \quad i \in k. \quad (19)$$

Assuming the posterior distribution $p(\theta|y_{(-k)})$ is summarized by S simulation draws $\theta^{k,s}$, we calculate its log predictive density as

$$\widehat{\text{elpd}}_i = \log \left(\frac{1}{S} \sum_{s=1}^S p(y_i|\theta^{k,s}) \right) \quad (20)$$

using the simulations corresponding to the subset k that contains data point i . We then sum to get the estimate

$$\widehat{\text{elpd}}_{\text{xval}} = \sum_{i=1}^n \widehat{\text{elpd}}_i. \quad (21)$$

There remains a bias as the model is learning from a fraction $\frac{1}{K}$ less of the data. Methods for correcting this bias exist but are rarely used as they can increase the variance, and if $K \geq 10$ the size of the bias is typically small compared to

the variance of the estimate (Vehtari and Lampinen 2002). In our experiments, exact LOO is the same as K -fold-CV with $K = N$ and we also analyze the effect of this bias and bias correction in Sect. 4.2.

For K -fold cross-validation, if the subjects are exchangeable, that is, the order does not contain information, then there is no need for random selection. If the order does contain information, e.g. in survival studies the later patients have shorter follow-ups, then randomization is often useful.

In most cases we recommend partitioning the data into subsets by randomly permuting the observations and then systemically dividing them into K subgroups. If the subjects are exchangeable, that is, the order does not contain information, then there is no need for random selection, but if the order does contain information, e.g. in survival studies the later patients have shorter follow-ups, then randomization is useful. In some cases it may be useful to stratify to obtain better balance among groups. See Vehtari and Lampinen (2002), Arolot and Celisse (2010), and Vehtari and Ojanen (2012) for further discussion of these points.

As the data can be divided in many ways into K groups it introduces additional variance in the estimates, which is also evident from our experiments. This variance can be reduced by repeating K -fold-CV several times with different permutations in the data division, but this will further increase the computational cost.

2.4 Data division

The purpose of using LOO or WAIC is to estimate the accuracy of the predictive distribution $p(\tilde{y}_i|y)$. Computation of PSIS-LOO and WAIC (and AIC and DIC) is based on computing terms $\log p(y_i|y) = \log \int p(y_i|\theta)p(\theta|y)$ assuming some agreed-upon division of the data y into individual data points y_i . Although often y_i will denote a single scalar observation, in the case of hierarchical data, it may denote a group of observations. For example, in cognitive or medical studies we may be interested in prediction for a new subject (or patient), and thus it is natural in cross-validation to consider an approach where y_i would denote all observations for a single subject and y_{-i} would denote the observations for all the other subjects. In theory, we can use PSIS-LOO and WAIC in this case, too, but as the number of observations per subject increases it is more likely that they will not work as well. The fact that importance sampling is difficult in higher dimensions is well known and is demonstrated for IS-LOO by Vehtari and Lampinen (2002) and for PSIS by Vehtari and Gelman (2015). The same problem can also be shown to hold for WAIC. If diagnostics warn about the reliability of PSIS-LOO (or WAIC), then K -fold cross-validation can be used by taking into account the hierarchical structure in the data when doing the data division as demonstrated, for example, by Vehtari and Lampinen (2002).

3 Implementation in Stan

We have set up code to implement LOO, WAIC, and K -fold cross-validation in R and Stan so that users will have a quick and convenient way to assess and compare model fits. Implementation is not automatic, though, because of the need to compute the separate factors $p(y_i|\theta)$ in the likelihood. Stan works with the joint density and in its usual computations does not “know” which parts come from the prior and which from the likelihood. Nor does Stan in general make use of any factorization of the likelihood into pieces corresponding to each data point. Thus, to compute these measures of predictive fit in Stan, the user needs to explicitly code the factors of the likelihood (actually, the terms of the log-likelihood) as a vector. We can then pull apart the separate terms and compute cross-validation and WAIC at the end, after all simulations have been collected. Sample code for carrying out this procedure using Stan and the `loo` R package is provided in Appendix. This code can be adapted to apply our procedure in other computing languages.

Although the implementation is not automatic when writing custom Stan programs, we can create implementations that are automatic for users of our new `rstanarm` R package (Gabry and Goodrich 2016). `rstanarm` provides a high-level interface to Stan that enables the user to specify many of the most common applied Bayesian regression models using standard R modeling syntax (e.g. like that of `glm`). The models are then estimated using Stan’s algorithms and the results are returned to the user in a form similar to the fitted model objects to which R users are accustomed. For the models implemented in `rstanarm`, we have preprogrammed many tasks, including computing and saving the pointwise predictive measures and importance ratios which we use to compute WAIC and PSIS-LOO. The `loo` method for `rstanarm` models requires no additional programming from the user after fitting a model, as we can compute all of the needed quantities internally from the contents of the fitted model object and then pass them to the functions in the `loo` package. Examples of using `loo` with `rstanarm` can be found in the `rstanarm` vignettes, and we also provide an example in Appendix 3 of this paper.

4 Examples

We illustrate with six simple examples: two examples from our earlier research in computing the effective number of parameters in a hierarchical model, three examples that were used by Epifani et al. (2008) to illustrate the estimation of the variance of the weight distribution, and one example of a multilevel regression from our earlier applied research. For each example we used the Stan default of 4 chains run for 1000 warmup and 1000 post-warmup iterations, yielding a

total of 4000 saved simulation draws. With Gibbs sampling or random-walk Metropolis, 4000 is not a large number of simulation draws. The algorithm used by Stan is Hamiltonian Monte Carlo with No-U-Turn-Sampling (Hoffman and Gelman 2014), which is much more efficient, and 1000 is already more than sufficient in many real-world settings. In these examples we followed standard practice and monitored convergence and effective sample sizes as recommended by Gelman et al. (2013). We performed 100 independent replications of all experiments to obtain estimates of variation. For the exact LOO results and convergence plots we run longer chains to obtain a total of 100,000 draws (except for the radon example which is much slower to run).

4.1 Example: Scaled 8 schools

For our first example we take an analysis of an education experiment used by Gelman et al. (2014) to demonstrate the use of information criteria for hierarchical Bayesian models.

The goal of the study was to measure the effects of a test preparation program conducted in eight different high schools in New Jersey. A separate randomized experiment was conducted in each school, and the administrators of each school implemented the program in their own way. Rubin (1981) performed a Bayesian meta-analysis, partially pooling the eight estimates toward a common mean. The model has the form, $y_i \sim N(\theta_i, \sigma_i^2)$ and $\theta_i \sim N(\mu, \tau^2)$, for $i = 1, \dots, n = 8$, with a uniform prior distribution on (μ, τ) . The measurements y_i and uncertainties σ_i are the estimates and standard errors from separate regressions performed for each school, as shown in Table 1. The test scores for the individual students are no longer available.

This model has eight parameters but they are constrained through their hierarchical distribution and are not estimated

Table 1 In a controlled study, independent randomized experiments were conducted in 8 different high schools to estimate the effect of special preparation for college admission tests

School	Estimated effect, y_j	Standard error of estimate, σ_j
A	28	15
B	8	10
C	−3	16
D	7	11
E	−1	9
F	1	11
G	18	10
H	12	18

Each row of the table gives an estimate and standard error from one of the schools. A hierarchical Bayesian model was fit to perform meta-analysis and use partial pooling to get more accurate estimates of the 8 effects. From Rubin (1981)

independently; thus we would anticipate the effective number of parameters should be some number between 1 and 8.

To better illustrate the behavior of LOO and WAIC, we repeat the analysis, rescaling the data points y by a factor ranging from 0.1 to 4 while keeping the standard errors σ unchanged. With a small data scaling factor the hierarchical model nears complete pooling and with a large data scaling factor the model approaches separate fits to the data for each school. Figure 1 shows $\widehat{\text{elpd}}$ for the various LOO approximation methods as a function of the scaling factor, based on 4000 simulation draws at each grid point.

When the data scaling factor is small (here, less than 1.5), all measures largely agree. As the data scaling factor increases and the model approaches no pooling, the population prior for θ_i gets flat and $p_{\text{waic}} \approx \frac{p}{2}$. This is correct behavior, as discussed by Gelman et al. (2014).

In the case of exact LOO, $\widehat{\text{lpd}} - \widehat{\text{elpd}}_{\text{loo}}$ can be larger than p . As the prior for θ_i approaches flatness, the log predictive density $p_{\text{post}(-i)}(y_i) \rightarrow -\infty$. At the same time, the full posterior becomes an inadequate approximation to $p_{\text{post}(-i)}$ and all approximations become poor approximations to the actual out-of-sample prediction error under the model.

WAIC starts to fail when one of the posterior variances of the log predictive densities exceeds 0.4. LOO approximations work well even if the tail shape k of the generalized Pareto distribution is between $\frac{1}{2}$ and 1, and the variance of the raw importance ratios is infinite. The error of LOO approximations increases with k , with a clearer difference between the methods when $k > 0.7$.

4.2 Example: Simulated 8 schools

In the previous example, we used exact LOO as the gold standard. In this section, we generate simulated data from the same statistical model and compare predictive performance on independent test data. Even when the number of observations n is fixed, as the scale of the population distribution increases we observe the effect of weak prior information in hierarchical models discussed in the previous section and by Gelman et al. (2014). Comparing the error, bias and variance of the various approximations, we find that PSIS-LOO offers the best balance.

For $i = 1, \dots, n = 8$, we simulate $\theta_{0,i} \sim N(\mu_0, \tau_0^2)$ and $y_i \sim N(\theta_{0,i}, \sigma_{0,i}^2)$, where we set $\sigma_{0,i} = 10$, $\mu_0 = 0$, and $\tau_0 \in \{1, 2, \dots, 30\}$. The simulated data is similar to the real 8 schools data, for which the empirical estimate is $\hat{\tau} \approx 10$. For each value of τ_0 we generate 100 training sets of size 8 and one test data set of size 1000. Posterior inference is based on 4000 draws for each constructed model.

Figure 2a shows the root mean square error (RMSE) for the various LOO approximation methods as a function of τ_0 , the scale of the population distribution. When τ_0 is large all of the approximations eventually have ever increasing

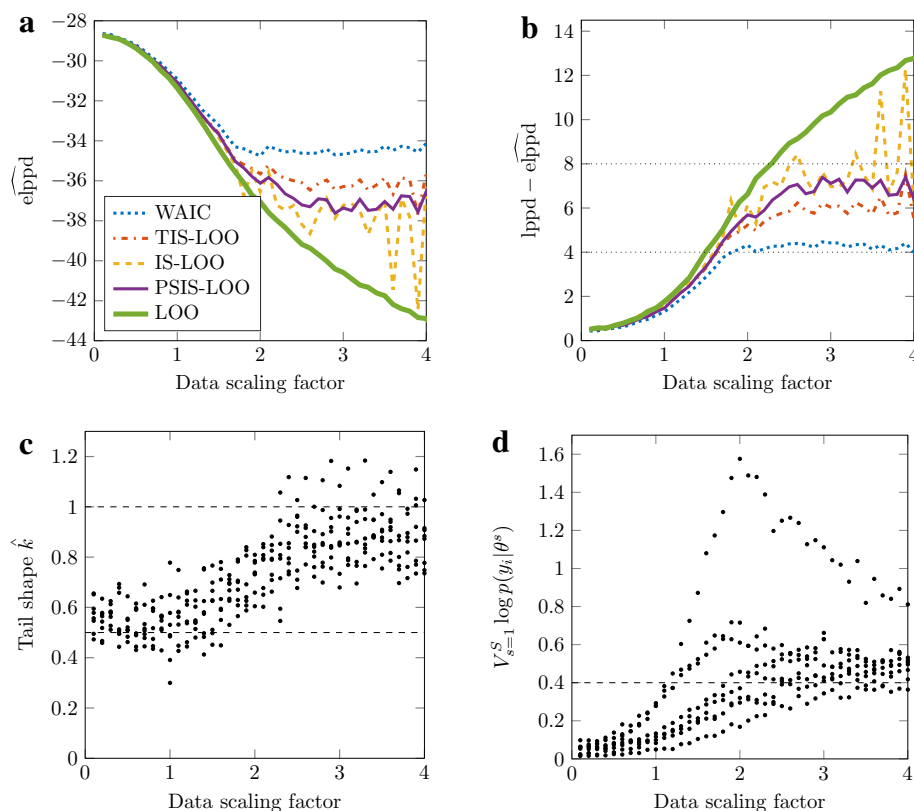


Fig. 1 8 Schools example: **a** WAIC, Truncated Importance Sampling LOO, Importance Sampling LOO, Pareto Smoothed Importance Sampling LOO, and exact LOO (which in this case corresponds to eightfold-CV); **b** estimated effective number of parameters for each of these measures; **c** tail shape $\hat{k}$ for the importance weights; and **d** the posterior variances of the log predictive densities, for scaled versions of the 8 schools data (the original observations y have been multiplied by a common factor). We consider scaling factors ranging from 0.1 (corresponding to near-zero variation of the underlying parameters among

the schools) to 4 (implying that the true effects in the schools vary by much more than their standard errors of measurement). As the scaling increases, eventually LOO approximations and WAIC fail to approximate exact LOO as the leave-one-out posteriors are not close to the full posterior. When the estimated tail shape $\hat{k}$ exceeds 1, the importance-weighted LOO approximations start to fail. When posterior variances of the log predictive densities exceeds 0.4, WAIC starts to fail. PSIS-LOO performs the best among the approximations considered here

RMSE, while exact LOO has an upper limit. For medium scales the approximations have *smaller* RMSE than exact LOO. As discussed later, this is explained by the difference in the variance of the estimates. For small scales WAIC has slightly smaller RMSE than the other methods (including exact LOO).

Watanabe (2010) shows that WAIC gives an asymptotically unbiased estimate of the out-of-sample prediction error—this does *not* hold for hierarchical models with weak prior information as shown by Gelman et al. (2014)—but exact LOO is slightly biased as the LOO posteriors use only $n - 1$ observations. WAIC's different behavior can be understood through the truncated Taylor series correction to the lpd, that is, not using the entire series will bias it towards lpd (see Sect. 2.2). The bias in LOO is negligible when n is large, but with small n it can be larger.

Figure 2b shows RMSE for the bias corrected LOO approximations using the first order correction of Burman (1989). For small scales the error of bias corrected LOOs is

smaller than WAIC. When the scale increases the RMSEs are close to the non-corrected versions. Although the bias correction is easy to compute, the difference in accuracy is negligible for most applications.

We shall discuss Fig. 2c in a moment, but first consider Fig. 3, which shows the RMSE of the approximation methods and the lpd of observed data decomposed into bias and standard deviation. All methods (except the lpd of observed data) have small biases and variances with small population distribution scales. Bias corrected exact LOO has practically zero bias for all scale values but the highest variance. When the scale increases the LOO approximations eventually fail and bias increases. As the approximations start to fail, there is a certain region where implicit shrinkage towards the lpd of observed data decelerates the increase in RMSE as the variance is reduced, even if the bias continues to grow.

If the goal were to minimize the RMSE for smaller and medium scales, we could also shrink exact LOO and increase

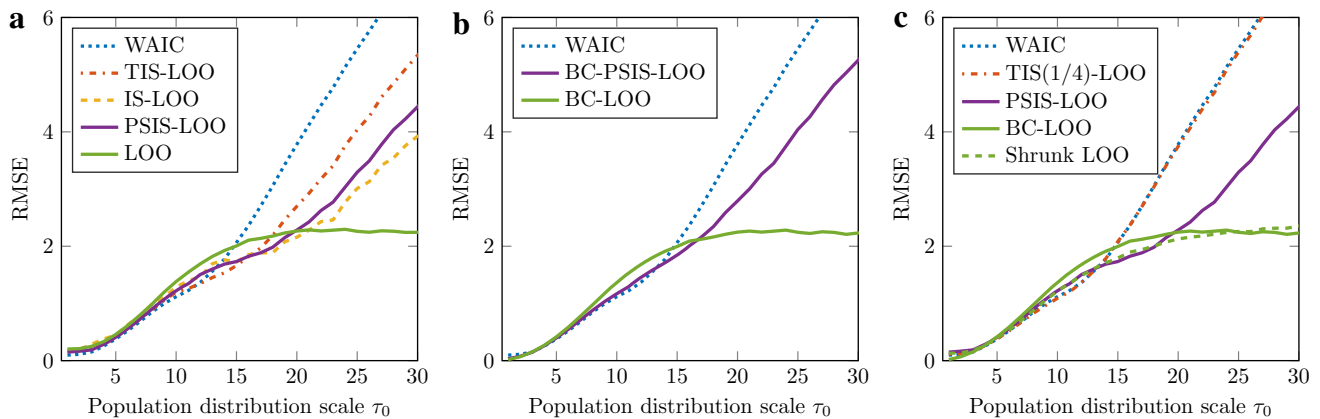


Fig. 2 Simulated 8 schools example: **a** Root mean square error of WAIC, Truncated Importance Sampling LOO, Importance Sampling LOO, Pareto Smoothed Importance Sampling LOO, and exact LOO with the true predictive performance computed using independently simulated test data; the error for all the methods increases, but the RMSE of exact LOO has an upper limit. Eventually the LOO approximations and WAIC fail to return exact LOO, as the leave-one-out posteriors are not close to the full posterior. When the estimated tail shape k exceeds 1, the importance-weighted LOO approximations start to fail. Among the approximations IS-LOO has the smallest RMSE as it has the smallest bias, and as the tail shape k is mostly below 1, it does not fail badly. **b** Root mean square error of WAIC, bias corrected Pareto Smoothed Importance Sampling LOO, and bias corrected exact LOO

with the true predictive performance computed using independently simulated test data. The bias correction also reduces RMSE, having the clearest impact with smaller population distribution scales, but overall the reduction in RMSE is negligible. **c** Root mean square error of WAIC, Truncated Importance Sampling LOO with heavy truncation ($\sqrt[4]{S\bar{r}}$), Pareto Smoothed Importance Sampling LOO, bias corrected exact LOO, and shrunk exact LOO with the true predictive performance computed using independently simulated test data. Truncated Importance Sampling LOO with heavy truncation matches WAIC accurately. Shrinking exact LOO towards the lpd of observed data reduces the RMSE for some scale values with small increase in error for larger scale values

shrinkage in approximations. Figure 2c shows the RMSE of the LOO approximations with two new choices. Truncated Importance Sampling LOO with very heavy truncation (to $\sqrt[4]{S\bar{r}}$) closely matches the performance of WAIC. In the experiments not included here, we also observed that adding more correct Taylor series terms to WAIC will make it behave similar to Truncated Importance Sampling with less truncation (see discussion of Taylor series expansion in Sect. 2.2). Shrunk exact LOO ($\alpha \cdot \text{elpd}_{\text{loo}} + (1 - \alpha) \cdot \text{lpd}$, with $\alpha = 0.85$ chosen by hand for illustrative purposes only) has a smaller RMSE for small and medium scale values as the variance is reduced, but the price is increased bias at larger scale values.

If the goal is robust estimation of predictive performance, then exact LOO is the best general choice because the error is limited even in the case of weak priors. Of the approximations, PSIS-LOO offers the best balance as well as diagnostics for identifying when it is likely failing.

4.3 Example: Linear regression for stack loss data

To check the performance of the proposed diagnostic for our second example we analyze the stack loss data used by [Peruggia \(1997\)](#) which is known to have analytically proven infinite variance of one of the importance weight distributions.

The data consist of $n = 21$ daily observations on one outcome and three predictors pertaining to a plant for the oxidation of ammonia to nitric acid. The outcome y is an inverse

measure of the efficiency of the plant and the three predictors x_1 , x_2 , and x_3 measure rate of operation, temperature of cooling water, and (a transformation of the) concentration of circulating acid.

[Peruggia \(1997\)](#) shows that the importance weights for leave-one-out cross-validation for the data point y_{21} have infinite variance. Figure 4 shows the distribution of the estimated tail shapes $\hat{k}$ and estimation errors compared to LOO in 100 independent Stan runs.³ The estimates of the tail shape $\hat{k}$ for $i = 21$ suggest that the variance of the raw importance ratios is infinite, however the generalized central limit theorem for stable distributions holds and we can still obtain an accurate estimate of the component of LOO for this data point using PSIS.

Figure 5 shows that if we continue sampling, the estimates for both the tail shape $\hat{k}$ and $\widehat{\text{elpd}}_i$ do converge (although slowly as $\hat{k}$ is close to 1). As the convergence is slow it would be more efficient to sample directly from $p(\theta^s | y_{-i})$ for the problematic i .

High estimates of the tail shape parameter $\hat{k}$ indicate that the full posterior is not a good importance sampling approximation to the desired leave-one-out posterior, and thus the observation is surprising according to the model. It is natural to consider an alternative model. We tried replacing the normal observation model with a Student- t to make the model

³ Smoothed density estimates were made using a logistic Gaussian process ([Vehtari and Riihimäki 2014](#)).

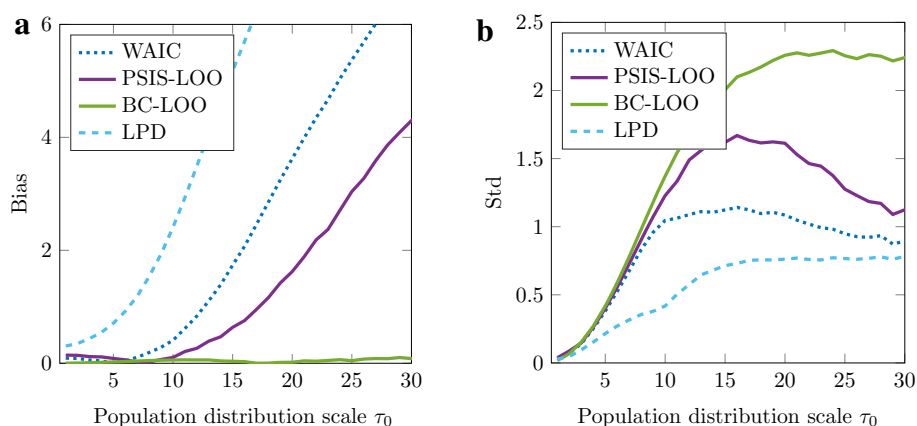


Fig. 3 Simulated 8 schools example: **a** Absolute bias of WAIC, Pareto Smoothed Importance Sampling LOO, bias corrected exact LOO, and the lpd (log predictive density) of observed data with the true predictive performance computed using independently simulated test data; **b** standard deviation for each of these measures; All methods except the lpd of observed data have small biases and variances with small pop-

ulation distribution scales. When the scale increases the bias of WAIC increases faster than the bias of the other methods (except the lpd of observed data). Bias corrected exact LOO has practically zero bias for all scale values. WAIC and Pareto Smoothed Importance Sampling LOO have lower variance than exact LOO, as they are shrunk towards the lpd of observed data, which has the smallest variance with all scales

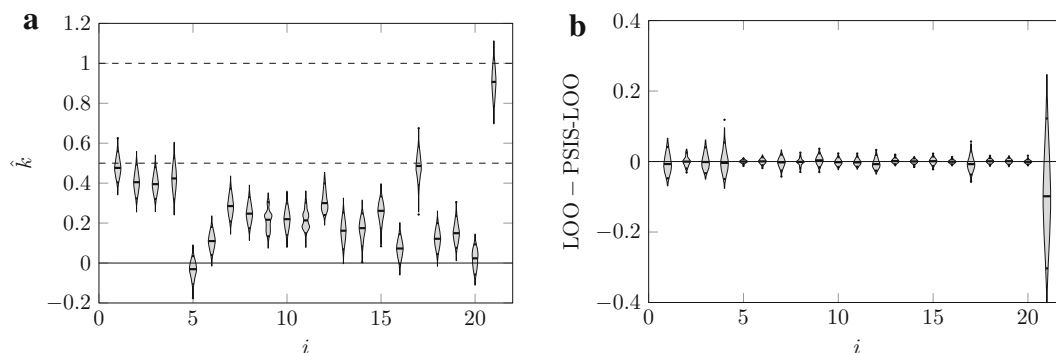


Fig. 4 Stack loss example with normal errors: Distributions of **a** tail shape estimates and **b** PSIS-LOO estimation errors compared to LOO, from 100 independent Stan runs. The pointwise calculation of the terms

in PSIS-LOO reveals that much of the uncertainty comes from a single data point, and it could make sense to simply re-fit the model to the subset and compute LOO directly for that point

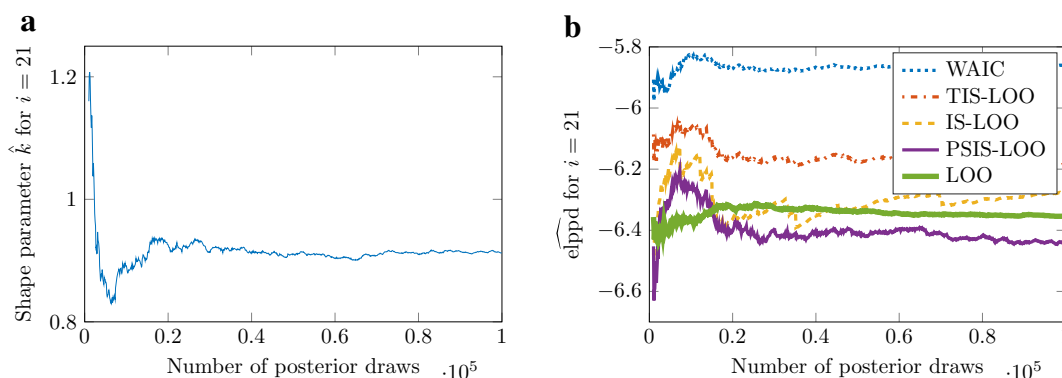


Fig. 5 Stack loss example with normal errors: **a** Tail shape estimate and **b** LOO approximations for the difficult point, $i = 21$. When more draws are obtained, the estimates converge (slowly) following the generalized central limit theorem

more robust for the possible outlier. Figure 6 shows the distribution of the estimated tail shapes $\hat{k}$ and estimation errors for PSIS-LOO compared to LOO in 100 independent Stan runs

for the Student- t linear regression model. The estimated tail shapes and the errors in computing this component of LOO are smaller than with Gaussian model.

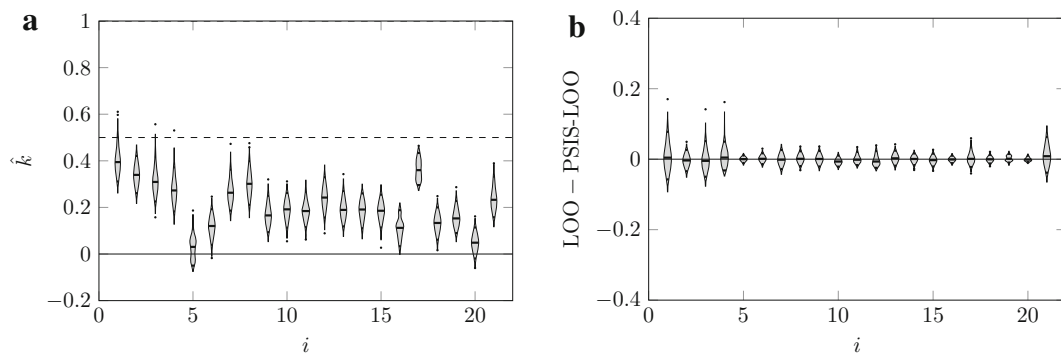


Fig. 6 Stack loss example with Student- t errors: Distributions of **a** tail shape estimates and **b** PSIS-LOO estimation errors compared to LOO, from 100 independent Stan runs. The computations are more stable than with normal errors (compare to Fig. 4)

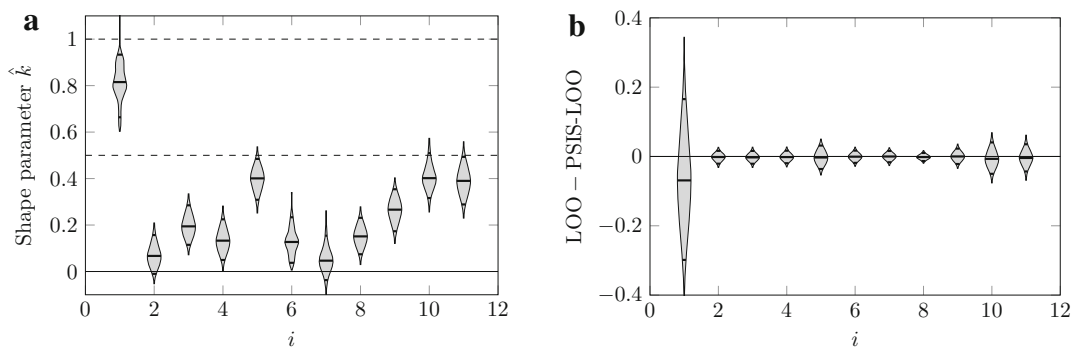


Fig. 7 Puromycin example: Distributions of **a** tail shape estimates and **b** PSIS-LOO estimation errors compared to LOO, from 100 independent Stan runs. In an applied example we would only perform these calculations once, but here we replicate 100 times to give a sense of the Monte Carlo error of our procedure

4.4 Example: Nonlinear regression for Puromycin reaction data

As a nonlinear regression example, we use the Puromycin biochemical reaction data also analyzed by Epifani et al. (2008). For a group of cells not treated with the drug Puromycin, there are $n = 11$ measurements of the initial velocity of a reaction, V_i , obtained when the concentration of the substrate was set at a given positive value, c_i . Velocity on concentration is given by the Michaelis-Menten relation, $V_i \sim N(mc_i/(\kappa + c_i), \sigma^2)$. Epifani et al. (2008) show that the raw importance ratios for observation $i = 1$ have infinite variance.

Figure 7 shows the distribution of the estimated tail shapes k and estimation errors compared to LOO in 100 independent Stan runs. The estimates of the tail shape k for $i = 1$ suggest that the variance of the raw importance ratios is infinite. However, the generalized central limit theorem for stable distributions still holds and we can get an accurate estimate of the corresponding term in LOO. We could obtain more draws to reduce the Monte Carlo error, or again consider a more robust model.

4.5 Example: Logistic regression for leukemia survival

Our next example uses a logistic regression model to predict survival of leukemia patients past 50 weeks from diagnosis. These data were also analyzed by Epifani et al. (2008). Explanatory variables are white blood cell count at diagnosis and whether “Auer rods and/or significant granulation of the leukemic cells in the bone marrow at diagnosis” were present.

Epifani et al. (2008) show that the raw importance ratios for data point $i = 15$ have infinite variance. Figure 8 shows the distribution of the estimated tail shapes k and estimation errors compared to LOO in 100 independent Stan runs. The estimates of the tail shape k for $i = 15$ suggest that the mean and variance of the raw importance ratios do not exist. Thus the generalized central limit theorem does not hold.

Figure 9 shows that if we continue sampling, the tail shape estimate stays above 1 and $\widehat{\text{elpd}}_i$ will not converge.

Large estimates for the tail shape parameter indicate that the full posterior is not a good importance sampling approximation for the desired leave-one-out posterior, and thus the observation is surprising. The original model used the white blood cell count directly as a predictor, and it would be natural

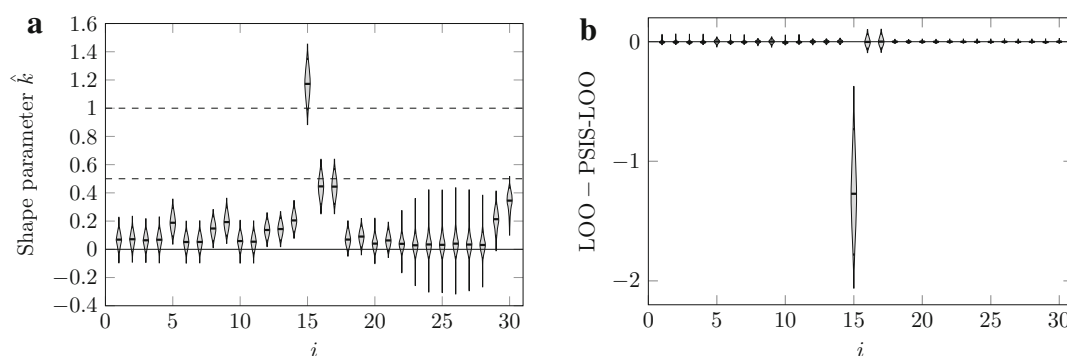


Fig. 8 Leukemia example: Distributions of **a** tail shape estimates and **b** PSIS-LOO estimation errors compared to LOO, from 100 independent Stan runs. The pointwise calculation of the terms in PSIS-LOO

reveals that much of the uncertainty comes from a single data point, and it could make sense to simply re-fit the model to the subset and compute LOO directly for that point

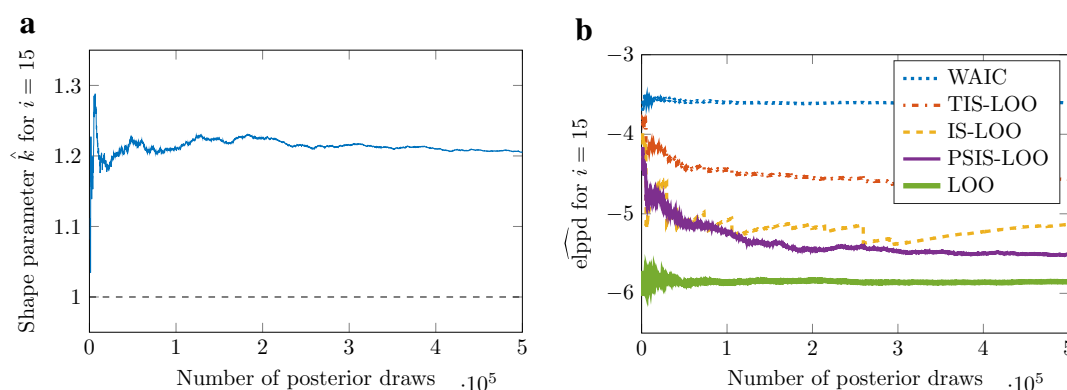


Fig. 9 Leukemia example: Distributions of **a** tail shape estimate and **b** LOO approximations for $i = 15$. If we continue sampling, the tail shape estimate stays above 1 and $\widehat{\text{elpd}}_i$ will not converge

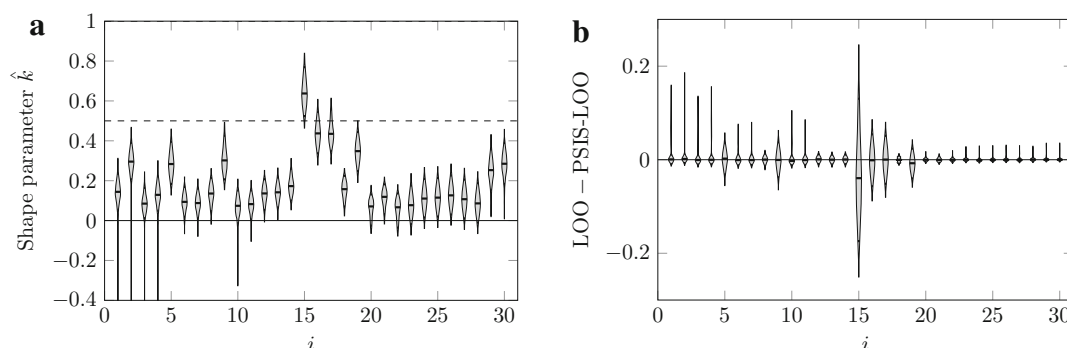


Fig. 10 Leukemia example with log-transformed predictor: **a** Distributions of tail shape estimates for each data point and **b** PSIS-LOO estimation errors compared to LOO, from 100 independent Stan runs. Computations are more stable compared to the model fit on the original scale and displayed in Fig. 8

to use its logarithm instead. Figure 10 shows the distribution of the estimated tail shapes k and estimation errors compared to LOO in 100 independent Stan runs for this modified model. Both the tail shape values and errors are now smaller.

4.6 Example: Multilevel regression for radon contamination

Gelman and Hill (2007) describe a study conducted by the United States Environmental Protection Agency designed to

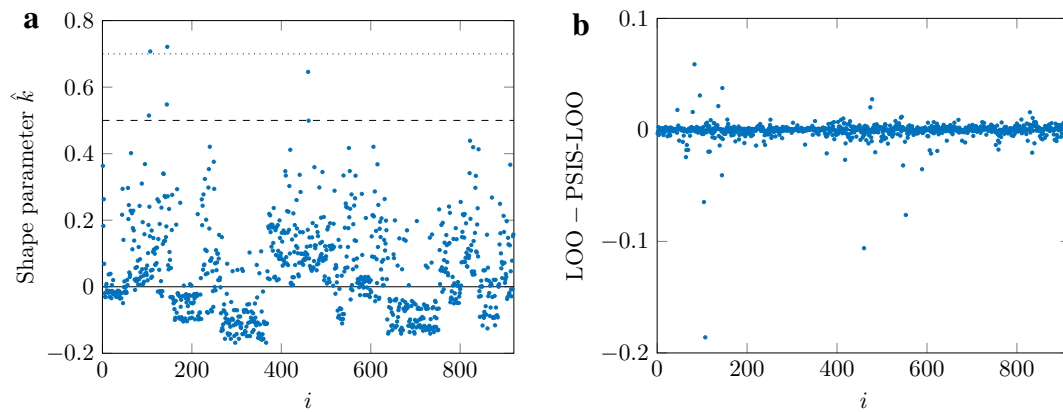


Fig. 11 Radon example: **a** Tail shape estimates for each point's contribution to LOO, and **b** error in PSIS-LOO accuracy for each data point, all based on a single fit of the model in Stan

measure levels of the carcinogen radon in houses throughout the United States. In high concentrations radon is known to cause lung cancer and is estimated to be responsible for several thousands of deaths every year in the United States. Here we focus on the sample of 919 houses in the state of Minnesota, which are distributed (unevenly) throughout 85 counties.

We fit the following multilevel linear model to the radon data

$$y_i \sim N(\alpha_{j[i]} + \beta_{j[i]}x_i, \sigma^2), \quad i = 1, \dots, 919$$

$$\begin{pmatrix} \alpha_j \\ \beta_j \end{pmatrix} \sim N\left(\begin{pmatrix} \gamma_0^\alpha + \gamma_1^\alpha u_j \\ \gamma_0^\beta + \gamma_1^\beta u_j \end{pmatrix}, \begin{pmatrix} \sigma_\alpha^2 & \rho\sigma_\alpha\sigma_\beta \\ \rho\sigma_\alpha\sigma_\beta & \sigma_\beta^2 \end{pmatrix}\right),$$

$$j = 1, \dots, 85,$$

where y_i is the logarithm of the radon measurement in the i th house, $x_i = 0$ for a measurement made in the basement and $x_i = 1$ if on the first floor (it is known that radon enters more easily when a house is built into the ground), and the county-level predictor u_j is the logarithm of the soil uranium level in the county. The residual standard deviation σ and all hyperparameters are given weakly informative priors. Code for fitting this model is provided in Appendix 3.

The sample size in this example ($n = 919$) is not huge but is large enough that it is important to have a computational method for LOO that is fast for each data point. Although the MCMC for the full posterior inference (using four parallel chains) finished in only 93 s, the computations for exact brute force LOO require fitting the model 919 times and took more than 20 h to complete (Macbook Pro, 2.6 GHz Intel Core i7). With the same hardware the PSIS-LOO computations took less than 5 s.

Figure 11 shows the results for the radon example and indeed the estimated shape parameters k are small and all of the tested methods are accurate. For two observations the

estimate of k is slightly higher than the preferred threshold of 0.7, but we can easily compute the elpd contributions for these points directly and then combine with the PSIS-LOO estimates for the remaining observations.⁴ This is the procedure we refer to as PSIS-LOO+ in Sect. 4.7 below.

4.7 Summary of examples

Table 2 compares the performance of Pareto smoothed importance sampling, raw importance sampling, truncated importance sampling, and WAIC for estimating expected out-of-sample prediction accuracy for each of the examples in Sects. 4.1–4.6. Models were fit in Stan to obtain 4000 simulation draws. In each case, the distributions come from 100 independent simulations of the entire fitting process, and the root mean squared error is evaluated by comparing to exact LOO, which was computed by separately fitting the model to each leave-one-out dataset for each example. The last three lines of Table 2 show additionally the performance of PSIS-LOO combined with direct sampling for the problematic i with $\hat{k} > 0.7$ (PSIS-LOO+), 10-fold-CV, and 10 times repeated 10-fold-CV.⁵ For the Stacks- N , Puromycin, and Leukemia examples, there was one i with $\hat{k} > 0.7$, and thus the improvement has the same computational cost as the full posterior inference. 10-fold-CV has higher RMSE than LOO approximations except in the Leukemia case. The higher RMSE of 10-fold-CV is due to additional variance from the data division. The repeated 10-fold-CV has smaller RMSE than basic 10-fold-CV, but now the cost of computation is already 100 times the original full posterior inference.

⁴ As expected, the two slightly high estimates for k correspond to particularly influential observations, in this case houses with extremely low radon measurements.

⁵ 10-fold-CV results were not computed for data sets with $n \leq 11$, and 10 times repeated 10-fold-CV was not feasible for the radon example due to the computation time required.

Table 2 Root mean square error for different computations of LOO as determined from a simulation study, in each case based on running Stan to obtain 4000 posterior draws and repeating 100 times

Method	8 schools	Stacks- N	Stacks- t	Puromycin	Leukemia	Leukemia-log	Radon
PSIS-LOO	0.21	0.21	0.12	0.20	1.33	0.18	0.34
IS-LOO	0.28	0.37	0.12	0.28	1.43	0.21	0.39
TIS-LOO	0.19	0.37	0.12	0.27	1.80	0.18	0.36
WAIC	0.40	0.68	0.12	0.46	2.30	0.29	1.30
PSIS-LOO+	0.21	0.11	0.12	0.10	0.11	0.18	0.34
10-fold-CV	—	1.34	1.01	—	1.62	1.40	2.87
10×10 -fold-CV	—	0.46	0.38	—	0.43	0.36	—

Methods compared are Pareto smoothed importance sampling (PSIS), PSIS with direct sampling if $\hat{k}_i > 0.7$ (PSIS-LOO+), raw importance sampling (IS), truncated importance sampling (TIS), WAIC, 10-fold-CV, and 10 times repeated 10-fold-CV for the different examples considered in Sects. 4.1–4.6: the hierarchical model for the 8 schools, the stack loss regression (with normal and t models), nonlinear regression for Puromycin, logistic regression for leukemia (in original and log scale), and hierarchical linear regression for radon. See text for explanations. PSIS-LOO and PSIS-LOO+ give the smallest error in all examples except the 8 schools, where it gives the second smallest error. In each case, we compared the estimates to the correct value of LOO by the brute-force procedure of fitting the model separately to each of the n possible training sets for each example

Table 3 Partial replication of Table 2 using 16,000 posterior draws in each case

Method	8 schools	Stacks- N	Stacks- t	Puromycin	Leukemia	Leukemia-log	Radon
PSIS-LOO	0.19	0.12	0.07	0.10	1.02	0.09	0.18
IS-LOO	0.13	0.21	0.07	0.25	1.21	0.11	0.24
TIS-LOO	0.15	0.27	0.07	0.17	1.60	0.09	0.24
WAIC	0.40	0.67	0.09	0.44	2.27	0.25	1.30

Monte Carlo errors are slightly lower. The errors for WAIC do not simply scale with $1/\sqrt{S}$ because most of its errors come from bias not variance

These results show that K -fold-CV is needed only if LOO approximations fail badly (see also the results in Vehtari and Lampinen 2002).

As measured by root mean squared error, PSIS consistently performs well. In general, when IS-LOO has problems it is because of the high variance of the raw importance weights, while TIS-LOO and WAIC have problems because of bias. Table 3 shows a replication using 16,000 Stan draws for each example. The results are similar results and PSIS-LOO is able to improve the most given additional draws.

5 Standard errors and model comparison

We next consider some approaches for assessing the uncertainty of cross-validation and WAIC estimates of prediction error. We present these methods in a separate section rather than in our main development because, as discussed below, the diagnostics can be difficult to interpret when the sample size is small.

5.1 Standard errors

The computed estimates $\widehat{\text{elpd}}_{\text{loo}}$ and $\widehat{\text{elpd}}_{\text{waic}}$ are each defined as the sum of n independent components so it is trivial to

compute their standard errors by computing the standard deviation of the n components and multiplying by $\sqrt{n}$. For example, define

$$\widehat{\text{elpd}}_{\text{loo},i} = \log p(y_i | y_{-i}), \quad (22)$$

so that $\widehat{\text{elpd}}_{\text{loo}}$ in (4) is the sum of these n independent terms. Then

$$\text{se}(\widehat{\text{elpd}}_{\text{loo}}) = \sqrt{n V_{i=1}^n \widehat{\text{elpd}}_{\text{loo},i}}, \quad (23)$$

and similarly for WAIC and K -fold cross-validation. The effective numbers of parameters, $\hat{p}_{\text{loo}}$ and $\hat{p}_{\text{waic}}$, are also sums of independent terms so we can compute their standard errors in the same way.

These standard errors come from considering the n data points as a sample from a larger population or, equivalently, as independent realizations of an error model. One can also compute Monte Carlo standard errors arising from the finite number of simulation draws using the formula from Gelman et al. (2013) which uses both between and within-chain information and is implemented in Stan. In practice we expect Monte Carlo standard errors to not be so interesting because we would hope to have enough simulations that the computations are stable, but it could make sense to look at them just

to check that they are low enough to be negligible compared to sampling error (which scales like $1/n$ rather than $1/S$).

The standard error (23) and the corresponding formula for $\text{se}(\widehat{\text{elpd}}_{\text{waic}})$ have two difficulties when the sample size is low. First, the n terms are not strictly independent because they are all computed from the same set of posterior simulations θ^s . This is a generic issue when evaluating the standard error of any cross-validated estimate. Second, the terms in any of these expressions can come from highly skewed distributions, so the second moment might not give a good summary of uncertainty. Both of these problems should subside as n becomes large. For small n , one could instead compute nonparametric error estimates using a Bayesian bootstrap on the computed log-likelihood values corresponding to the n data points (Vehtari and Lampinen 2002).

5.2 Model comparison

When comparing two fitted models, we can estimate the difference in their expected predictive accuracy by the difference in $\widehat{\text{elpd}}_{\text{loo}}$ or $\widehat{\text{elpd}}_{\text{waic}}$ (multiplied by -2 , if desired, to be on the deviance scale). To compute the standard error of this difference we can use a paired estimate to take advantage of the fact that the same set of n data points is being used to fit both models.

For example, suppose we are comparing models A and B, with corresponding fit measures $\widehat{\text{elpd}}_{\text{loo}}^A = \sum_{i=1}^n \widehat{\text{elpd}}_{\text{loo},i}^A$ and $\widehat{\text{elpd}}_{\text{loo}}^B = \sum_{i=1}^n \widehat{\text{elpd}}_{\text{loo},i}^B$. The standard error of their difference is simply,

$$\text{se}(\widehat{\text{elpd}}_{\text{loo}}^A - \widehat{\text{elpd}}_{\text{loo}}^B) = \sqrt{n \sum_{i=1}^n (\widehat{\text{elpd}}_{\text{loo},i}^A - \widehat{\text{elpd}}_{\text{loo},i}^B)^2}, \quad (24)$$

and similarly for WAIC and K -fold cross-validation. Alternatively the non-parametric Bayesian bootstrap approach can be used (Vehtari and Lampinen 2002).

As before, these calculations should be most useful when n is large, because then non-normality of the distribution is not such an issue when estimating the uncertainty of these sums.

In any case, we suspect that these standard error formulas, for all their flaws, should give a better sense of uncertainty than what is obtained using the current standard approach for comparing differences of deviances to a χ^2 distribution, a practice that is derived for Gaussian linear models or asymptotically and, in any case, only applies to nested models.

Further research needs to be done to evaluate the performance in model comparison of (24) and the corresponding standard error formula for LOO. Cross-validation and WAIC should not be used to select a single model among a

large number of models due to a selection induced bias as demonstrated, for example, by Piironen and Vehtari (2016).

We demonstrate the practical use of LOO in model comparison using the radon example from Sect. 4.6. Model A is the multilevel linear model discussed in Sect. 4.6 and Model B is the same model but without the county-level uranium predictor. That is, at the county-level Model B has

$$\begin{pmatrix} \alpha_j \\ \beta_j \end{pmatrix} \sim N \left(\begin{pmatrix} \mu_\alpha \\ \mu_\beta \end{pmatrix}, \begin{pmatrix} \sigma_\alpha^2 & \rho\sigma_\alpha\sigma_\beta \\ \rho\sigma_\alpha\sigma_\beta & \sigma_\beta^2 \end{pmatrix} \right), \quad j = 1, \dots, 85.$$

Comparing the models on PSIS-LOO reveals an estimated difference in elpd of 10.2 (with a standard error of 5.1) in favor of Model A.

5.3 Model comparison using pointwise prediction errors

We can also compare models in their leave-one-out errors, point by point. We illustrate with an analysis of a survey of residents from a small area in Bangladesh that was affected by arsenic in drinking water. Respondents with elevated arsenic levels in their wells were asked if they were interested in getting water from a neighbor's well, and a series of models were fit to predict this binary response given various information about the households (Gelman and Hill 2007).

Here we start with a logistic regression for the well-switching response given two predictors: the arsenic level of the water in the resident's home, and the distance of the house from the nearest safe well. We compare this to an alternative logistic regression with the arsenic predictor on the logarithmic scale. The two models have the same number of parameters but give different predictions.

Figure 12 shows the pointwise results for the arsenic example. The scattered blue dots on the left side of Fig. 12a and on the lower right of Fig. 12b correspond to data points which Model A fits particularly poorly—that is, large negative contributions to the expected log predictive density. We can also sum these n terms to yield an estimated difference in elpd_{loo} of 16.4 with a standard error of 4.4. This standard error derives from the finite sample size and is scaled by the variation in the differences displayed in Fig. 12; it is *not* a Monte Carlo error and does not decline to 0 as the number of Stan simulation draws increases.

6 Discussion

This paper has focused on the practicalities of implementing LOO, WAIC, and K -fold cross-validation within a Bayesian simulation environment, in particular the coding of the log-likelihood in the model, the computations of the information

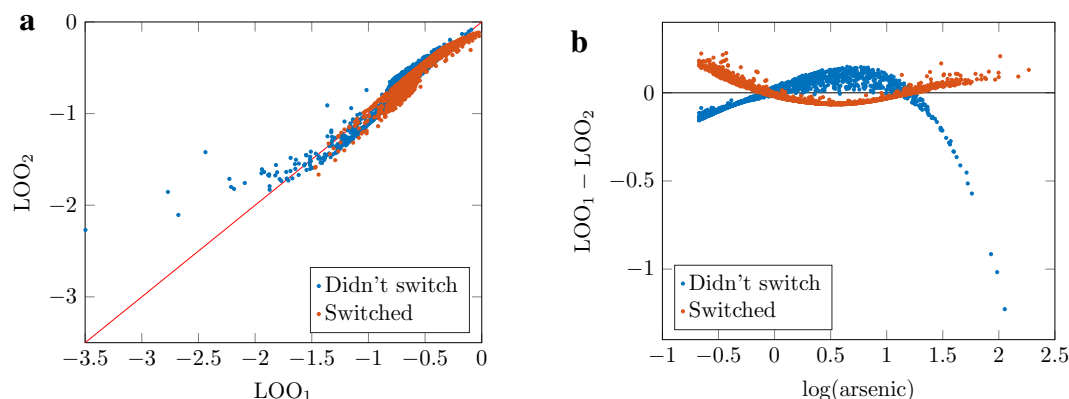


Fig. 12 Arsenic example, comparing two models in terms of their pointwise contributions to LOO: **a** comparing contributions of LOO directly; **b** plotting the difference in LOO as a function of a key predictor (the existing arsenic level). To aid insight, we have colored the data according to the (binary) output, with red corresponding to $y = 1$

measures, and the stabilization of weights to enable an approximation of LOO without requiring refitting the model.

Some difficulties persist, however. As discussed above, any predictive accuracy measure involves two definitions: (1) the choice of what part of the model to label as “the likelihood”, which is directly connected to which potential replications are being considered for out-of-sample prediction; and (2) the factorization of the likelihood into “data points”, which is reflected in the later calculations of expected log predictive density.

Some choices of replication can seem natural for a particular dataset but less so in other comparable settings. For example, the 8 schools data are available only at the school level and so it seems natural to treat the school-level estimates as data. But if the original data had been available, we would surely have defined the likelihood based on the individual students’ test scores. It is an awkward feature of predictive error measures that they might be determined based on computational convenience or data availability rather than fundamental features of the problem. To put it another way, we are assessing the fit of the model to the particular data at hand.

Finally, these methods all have limitations. The concern with WAIC is that formula (12) is an asymptotic expression for the bias of lpd for estimating out-of-sample prediction error and is only an approximation for finite samples. Cross-validation (whether calculated directly by re-fitting the model to several different data subsets, or approximated using importance sampling as we did for LOO) has a different problem in that it relies on inference from a smaller subset of the data being close to inference from the full dataset, an assumption that is typically but not always true.

For example, as we demonstrated in Sect. 4.1, in a hierarchical model with only one data point per group, PSIS-LOO

and blue representing $y = 0$. For any given data point, one model will fit better than another, but for this example the graphs reveal that the difference in LOO between the models arises from the linear model’s poor predictions for 10–15 non-switchers with high arsenic levels

and WAIC can dramatically understate prediction accuracy. Another setting where LOO (and cross-validation more generally) can fail is in models with weak priors and sparse data. For example, consider logistic regression with flat priors on the coefficients and data that happen to be so close to separation that the removal of a single data point can induce separation and thus infinite parameter estimates. In this case the LOO estimate of average prediction accuracy will be zero (that is, $\text{elpd}_{\text{is-loo}}$ will be $-\infty$) if it is calculated to full precision, even though predictions of future data from the actual fitted model will have bounded loss. Such problems should not arise asymptotically with a fixed model and increasing sample size but can occur with actual finite data, especially in settings where models are increasing in complexity and are insufficiently constrained.

That said, quick estimates of out-of-sample prediction error can be valuable for summarizing and comparing models, as can be seen from the popularity of AIC and DIC. For Bayesian models, we prefer PSIS-LOO and K-fold cross-validation to those approximations which are based on point estimation.

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Appendix: Implementation in Stan and R

Appendix 1: Stan code for computing and storing the pointwise log-likelihood

We illustrate how to write Stan code that computes and stores the pointwise log-likelihood using the arsenic example from Sect. 5.3. We save the program in the file `logistic.stan`:

```

data {
  int N;
  int P;
  int<lower=0,upper=1> y[N];
  matrix[N,P] X;
}
parameters {
  vector[P] b;
}
model {
  b ~ normal(0,1);
  y ~ bernoulli_logit(X*b);
}
generated quantities {
  vector[N] log_lik;
  for (n in 1:N)
    log_lik[n] = bernoulli_logit_lpmf(y[n] | X[n]*b);
}

```

We have defined the log-likelihood as a vector `log_lik` in the generated quantities block so that the individual terms will be saved by Stan.⁶ It would seem desirable to compute the terms of the log-likelihood directly without requiring the repetition of code, perhaps by flagging the appropriate lines in the model or by identifying the log likelihood as those lines in the model that are defined relative to the data. But there are so many ways of writing any model in Stan—anything goes as long as it produces the correct log posterior density, up to any arbitrary constant—that we cannot see any general way at this time for computing LOO and WAIC without repeating the likelihood part of the code. The good news is that the additional computations are relatively cheap: sitting as they do in the generated quantities block (rather than in the transformed parameters and model blocks), the expressions for the terms of the log posterior need only be computed once per saved iteration rather than once per HMC leapfrog step, and no gradient calculations are required.

Appendix 2: The `loo` R package for LOO and WAIC

The `loo` R package provides the functions `loo()` and `waic()` for efficiently computing PSIS-LOO and WAIC for fitted Bayesian models using the methods described in this paper.

These functions take as their argument an $S \times n$ log-likelihood matrix, where S is the size of the posterior sample (the number of retained draws) and n is the number of data points.⁷ The required means and variances across simula-

tions are calculated and then used to compute the effective number of parameters and LOO or WAIC.

The `loo()` function returns $\widehat{\text{elpd}}_{\text{loo}}$, $\widehat{p}_{\text{loo}}$, $\text{looic} = -2 \widehat{\text{elpd}}_{\text{loo}}$ (to provide the output on the conventional scale of “deviance” or AIC),⁸ the pointwise contributions of each of these measures, and standard errors. The `waic()` function computes the analogous quantities for WAIC. Also returned by the `loo()` function is the estimated shape parameter $\hat{k}$ for the generalized Pareto fit to the importance ratios for each leave-one-out distribution. These computations could also be implemented directly in Stan C++, perhaps following the rule that the calculations are performed if there is a variable named `log_lik`. The `loo` R package, however, is more general and does not require that a model be fit using Stan, as long as an appropriate log-likelihood matrix is supplied.

Using the `loo` package. Below, we provide R code for preparing and running the logistic regression for the arsenic example in Stan. After fitting the model we then use the `loo` package to compute LOO and WAIC.⁹

```

library("rstan")
library("loo")
# Read in and prepare the data
wells <- read.csv("wells.csv")
N <- nrow(wells)
X <- cbind(rep(1,N), wells$dist100, wells$arsenic)
y <- wells$y
P <- ncol(X)
# Fit the model with Stan
fit_1 <- stan("logistic.stan")
print(fit_1, "b")
# Compute LOO
log_lik_1 <- extract_log_lik(fit_1)
loo_1 <- loo(log_lik_1)
print(loo_1)

```

The printed output shows $\widehat{\text{elpd}}_{\text{loo}}$, $\widehat{p}_{\text{loo}}$, looic , and their standard errors:

```

Computed from 4000 by 3020 log-likelihood matrix
      Estimate   SE
elpd_loo -1968.3 15.6
p_loo      3.1  0.1
looic     3936.6 31.2
All Pareto k estimates OK (k < 0.5)

```

By default, the estimates for the shape parameter $\hat{k}$ of the generalized Pareto distribution are also checked and a message is displayed informing the user if any $\hat{k}$ are problematic (see the end of Sect. 2.1). In the example above the message

⁶ The code in the generated quantities block is written using the new syntax introduced in Stan version 2.10.0.

⁷ For models fit to large datasets it can be infeasible to store the entire log-likelihood matrix in memory. A function for computing the log-likelihood from the data and posterior draws of the relevant parameters may be specified instead of the log-likelihood matrix—the necessary data and draws are supplied as an additional argument—and columns of the log-likelihood matrix are computed as needed. This requires less memory than storing the entire log-likelihood matrix and allows `loo` to be used with much larger datasets.

⁸ In statistics there is a tradition of looking at deviance, while in computer science the log score is more popular, so we return both.

⁹ The `extract_log_lik()` function used in the example is a convenience function for extracting the log-likelihood matrix from a fitted Stan model provided that the user has computed and stored the pointwise log-likelihood in their Stan program (see, for example, the generated quantities block in 1). The argument `parameter_name` (defaulting to “`log_lik`”) can also be supplied to indicate which parameter or generated quantity corresponds to the log-likelihood.

tells us that all of the estimates for $\hat{k}$ are fine. However, if any $\hat{k}$ were between 1/2 and 1 or greater than 1 the message would instead look something like this:

Warning messages:

```
1: 200 (7%) Pareto k estimates between 0.5 and 1
2: 85 (3%) Pareto k estimates greater than 1
```

If there are any warnings then it can be useful to visualize the estimates to check which data points correspond to the large $\hat{k}$ values. A plot of the $\hat{k}$ estimates can also be generated using `plot(loo1)` and the list returned by the `loo` function also contains the full vector of $\hat{k}$ values.

Model comparison To compare this model to a second model on their values of LOO we can use the `compare` function:

```
# First fit a second model, using log(arsenic)
# instead of arsenic
X <- cbind(rep(1,N), wells$dist100,
  log(wells$arsenic))
P <- ncol(X)
fit_2 <- stan("logistic.stan")
print(fit_2, "b")
log_lik_2 <- extract_log_lik(fit_2)
loo_2 <- loo(log_lik_2)

# Compare the models
loo_diff <- compare(loo_1, loo_2)
print(loo_diff)
```

This new object, `loo_diff`, contains the estimated difference of expected leave-one-out prediction errors between the two models, along with the standard error:

```
elpd_diff      SE
      16.1      4.4
```

Code for WAIC For WAIC the code is analogous and the objects returned have the same structure (except there are no Pareto k estimates). The `compare()` function can also be used to estimate the difference in WAIC between two models:

```
waic_1 <- waic(log_lik_1)
waic_2 <- waic(log_lik_2)
waic_diff <- compare(waic_1, waic_2)
```

Appendix 3: Using the `loo` R package with `rstanarm` models

Here we show how to fit the model for the radon example from Sect. 4.6 and carry out PSIS-LOO using the `rstanarm` and `loo` packages.

```
library("rstanarm")
# The subset of the radon data we need is included
# in rstanarm
data(radon)
# Fit the first model
modelA <- stan_lmer(
  log_radon ~ floor + log_uranium + floor:log_uranium
  + (1 + floor | county),
  data = radon,
  cores = 4,
```

```
  iter = 2000,
  chains = 4
)
# Fit the model without the county-level
# uranium predictor
modelB <- update(fitA, formula = log_radon ~
  floor + (1 + floor | county))
```

After fitting the models we can pass the fitted model objects `modelA` and `modelB` directly to `rstanarm`'s `loo` method and it will call the necessary functions from the `loo` package internally.

```
# Compare models
looA <- loo(modelA)
looB <- loo(modelB)
compare(looA, looB)
```

This returns:

```
elpd_diff      se
      -10.2      5.2
```

If there are warnings about large values of the estimated Pareto shape parameter $\hat{k}$ for the importance ratios, `rstanarm` is also able to automatically carry out the procedure we call PSIS-LOO+ (see Sect. 4.7). That is, `rstanarm` can refit the model, leaving out these problematic observations one at a time and computing their elpd contributions directly. Then these values are combined with the results from PSIS-LOO for the other observations and returned to the user. We recommend this when there are only a few large $\hat{k}$ estimates. If there are many of them then we recommend K -fold cross-validation, which is also implemented in the latest release of `rstanarm`.

Appendix 4: Stan code for K -fold cross-validation

To implement K -fold cross-validation we repeatedly partition the data, with each partition fitting the model to the training set and using it to predict the holdout set. The code for cross-validation does not look so generic because of the need to repeatedly partition the data. However, in any particular example the calculations are not difficult to implement, the main challenge being the increase in computation time by roughly a factor of K . We recommend doing the partitioning in R (or Python, or whichever data-processing environment is being used) and then passing the training data and holdout data to Stan in two pieces.

Again we illustrate with the logistic regression for the arsenic example. We start with the model from above, but we pass in both the training data (N_t , y_t , X_t) and the holdout set (N_h , y_h , X_h), augmenting the data block accordingly. We then alter the generated quantities block to operate on the holdout data:


```

data {
  int P;                      // Number of regression predictors
  int N_t;                    // (Training) number of data points
  int<lower=0,upper=1> y_t[N_t]; // (Training) binary data
  matrix[N_t,P] X_t;         // (Training) predictors
  int N_h;                    // (Holdout)
  int y_h[N_h];              // (Holdout)
  matrix[N_h,P] X_h;         // (Holdout)
  real a;
}
parameters {
  vector[P] b;
}
model {
  y_t ~ bernoulli_logit(X_t*b);
}
generated quantities {
  vector[N_t] log_lik_t;
  vector[N_h] log_lik_h;
  for (n in 1:N_t)
    log_lik_t[n] = bernoulli_logit_lpmf(y_t[n] | X_t[n]*b);
  for (n in 1:N_h)
    log_lik_h[n] = bernoulli_logit_lpmf(y_h[n] | X_h[n]*b);
}

```

LOO could be also implemented in this way, setting N_t to $N-1$ and N_h to 1. But, as discussed in the article, for large datasets it is more practical to approximate LOO using importance sampling on the draws from the posterior distribution fit to the entire dataset.

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